

 Adjustment for operation with heavy load

9. Load operating test

B1. Low-frequency loading performance is below 1/10 of rated frequency:

- a. If the low-frequency loading performance is insufficient, or the rotor speed is not smooth, increase Pr.10-31 (current command of I/F mode).
- b. If the low-frequency current is large, decrease Pr.10-31 (current command of I/F mode).

B2. Test the with-load accelerating performance:

When the motor operates in 1/10 of rotor speed and above, if the speed cannot follow the acceleration time during accelerating, or the current stalls, increase Pr.07-38 (PMSVC voltage feedback forward gain).

10. Stability test at constant speed operation: the motor operates stably at constant speed

- a. If the motor output rotor speed presents periodic low-frequency wave, increase Pr.10-34 (PM sensorless speed estimator low-pass filter gain), or increase Pr.10-32 (PM FOC sensorless speed estimator bandwidth).
- b. If the output frequency reflects high frequency vibration, decrease Pr.10-34 or decrease Pr.10-32.

● PM SVC related parameters

Refer to Section 12-1 Description of Parameter Settings for more details.

Parameter	Description	Unit	Default	Setting Range
Pr.07-24	Torque command filter time	sec.	0.050	0.001–10.000
Pr.07-26	Torque compensation gain	NA	0	0–5000
Pr.07-38	PMSVC voltage feedback forward gain	NA	1.0	0.00–2.00
Pr.10-31	I/F mode, current command	%	40	0–150
Pr.10-32	PM FOC sensorless speed estimator bandwidth	Hz	5.00	0.00–600.00
Pr.10-34	PM sensorless speed estimator low-pass filter gain	NA	1.00	0.00–655.35
Pr.10-39	Frequency point to switch from I/F mode to PM sensorless mode	Hz	20.00	0.00–599.00
Initial Angle Estimating Parameters				
Pr.10-42	Initial angle detection pulse value	NA	1.0	0.0–3.0
Pr.10-51	Injection frequency	Hz	500	0–1200
Pr.10-52	Injection magnitude	V	15.0 / 30.0	0.0–200.0
Pr.10-53	PM initial rotor position detection method 0: Disable 1: Force attracting the rotor to zero degrees 2: High frequency injection 3: Pulse injection	NA	0	0–3